

Collinearity

Problem 1. OA, OB, OC are three chords of a circle. The circles with diameters OA, OB meet again at Z , the circles with diameters OB, OC meet again at X , and the circles with diameters OC, OA meet again at Y . Show that X, Y, Z are collinear.

Problem 2. Let H be the orthocentre of an acute $\triangle ABC$. The tangents AP, AQ are constructed from A to the circle with diameter BC . The contact points are P and Q respectively. Prove that P, H, Q are collinear.

Problem 3. Let P be a point inside $\triangle ABC$. The line through P perpendicular to AP meets the sideline BC at X . The line through P perpendicular to BP meets the sideline CA at Y . The line through P perpendicular to CP meets the sideline AB at Z . Prove that X, Y, Z are collinear.

Problem 4. Let H be the orthocentre of an acute triangle ABC . Prove that the midpoints of AB and CH and the intersection point of the angle bisectors of $\angle CAH$ and $\angle CBH$ lie on the same line.

Problem 5. Let AB be a diameter of a circle Γ and let O be the centre of Γ . Let C be an arbitrary point on Γ . The tangents at A and C to Γ intersect at a point D . A circle ω passing through C is tangent to the line AD at D . Let DE be a diameter of ω . The line BD meets ω again at F . Let M be the midpoint of OA . Prove that E, F, M are collinear.

Problem 6. Given acute $\triangle ABC$, let P be a point on the perpendicular bisector of the side BC such that P and A lie on the same side of the line BC . Let Q be a point on the opposite side of the line BC as P such that $\angle BCQ = \angle ABP$ and $\angle CBQ = \angle ACP$. Prove that A, P, Q are collinear.

Problem 7. Let I be the incentre of $\triangle ABC$. Let D be the contact point of the incircle of $\triangle ABC$ and the side BC . Denote by H_1 and H_2 the orthocentres of $\triangle AIB$ and $\triangle AIC$ respectively. Prove that H_1, H_2, D are collinear.

Problem 8. Let H be the orthocentre of an acute triangle ABC and M the midpoint of side BC . If D and E are the feet of perpendicular of H on the internal and external angle bisectors of $\angle BAC$, prove that M, D and E are collinear.

Problem 9. Let O and H be the circumcentre and orthocentre of $\triangle ABC$ respectively. Let D be the foot of the altitude from A . The line through D perpendicular to OD meets the extension of AC at E . The circumcircle of $\triangle CAH$ intersects the line BC again at F . Prove that H, E, F are collinear.

Problem 10. In $\triangle ABC$, $AB > AC$. Let M be the midpoint of BC . The perpendicular bisector of BC intersects the circumcircle of $\triangle ABC$ at the points D and E , where D lies on the same side of BC as A . The circumcircle of $\triangle AME$ meets the line AB and AC again at P and Q respectively. The circumcircles of $\triangle ACP$ and $\triangle ABQ$ meet again at X . Prove that D, A, X are collinear.